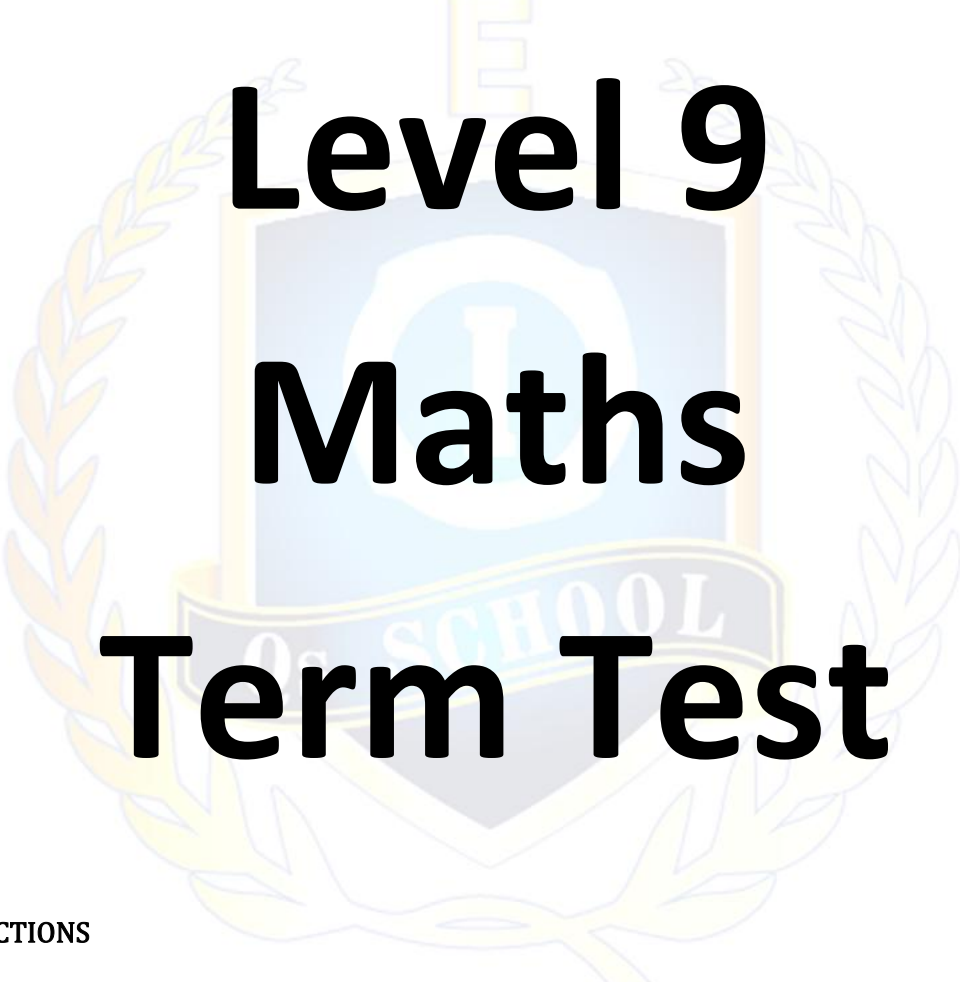


Student ID: _____

Student Name: _____

Teacher: _____



Level 9 Maths Term Test

INSTRUCTIONS

1. Write your Student ID & Name on the on top of the page.
2. Please answer all questions (**Total marks 50**).
3. You have **45 minutes** of writing time.
4. If you need the help of a supervisor during the test, please raise your hand.
5. No calculator permitted.

DO NOT OPEN THIS TEST PAPER UNTIL YOU ARE TOLD.

Short-answer questions**Question 1**

Expand the following expressions.

a) $(g - 8)^2$

b) $(6y + 7)^2$

c) $(7d - 3w)^2$

d) $\left(\frac{x}{2} - 2y\right)^2$

(1 × 4 = 4 marks)

Question 2

Factorise the following completely.

a) $x^2 + 2x - 15$

b) $x^2 - 9x + 20$

c) $4x^2 + 4x - 24$

d) $xy + 2x + 7y + 14$

e) $z^2 - 3z - 5z + 15$

f) $2x^4 - 32y^4$

(1 × 6 = 6 marks)

Question 3

The product of three numbers is $5x^2 + 5x - 210$. One of the numbers is 5. Find the other two numbers, expressing them in terms of x in the form $(x \pm a)$ and $(x \pm b)$.

(2 marks)

Question 4

Sketch the following parabolas, showing the axis intercepts and the turning point in full coordinate form.

a) $y = -x^2 + 2x + 15$

b) $y = 3x^2 + 9x$

(2 × 3 = 6 marks)

Question 5

A book cover has an area of 60cm^2 . If the width is 7 cm less than the length, find the dimensions of the book cover.

(2 marks)

Question 6

Solve the following equations.

a) $\frac{2k + 3}{2} + \frac{5 - k}{3} = 2$

b) $\frac{2x + 3}{4} - \frac{3x - 5}{3} = 2$

(2 × 2 = 4 marks)

Question 7

Solve the following inequalities.

a) $\frac{3x - 15}{18} \geq \frac{9 - x}{4}$

b) $\frac{1}{2} > \frac{2x}{5} + \frac{6 - 3x}{9}$

(2 × 2 = 4 marks)

Question 8

Find the values of x and y for each of the following questions.

a)

$$3x + 2y = -6$$

$$2x + 4y = -20$$

b)

$$2x + y - 2 = 0$$

$$-3x - 4y = 4.5$$

(2 × 3 = 6 marks)

Question 9

A toothbrush and a razor cost \$1.10 in total. The razor costs \$1 more than the toothbrush.

How much does each item cost?

(2 marks)

Question 10

The average of two positive integers is 34. The sum of the larger number and double the smaller number is 88. Find the two numbers.

(2 marks)

Question 11

A book is opened and the product of the two page numbers showing is 342. Find these two current page numbers. Also, state the larger of the two page numbers showing when the book is turned to the next page.

(3 marks)

Extended Response Question**Question 1**

Leonard and Faye want to develop a vegetable garden in their backyard. They have 20 metres of chicken wire and enough posts to erect a fence around the four sides of the rectangular garden.

- a) If the length of the rectangular garden is x m, find an expression (in terms of x) for the width of the garden. Hint – draw a birds-eye view of the garden and label the dimensions.

- b) Hence, show that the area of the garden, $A \text{ m}^2$, is given by the quadratic expression:

$$A = -x^2 + 10x$$

- c) Sketch the graph of the area of the vegetable garden, $A \text{ m}^2$, against the length of the garden, $x \text{ m}$, in the appropriate regions. Display all relevant axis intercepts and the turning point in full coordinate form. Show length on the horizontal axis.
- d) What should be the dimensions (i.e. the length and the width) of the garden if the area is to be as large as possible?
- e) What is the area of the largest vegetable garden that Leonard and Faye can possibly make?

(1 + 2 + 3 + 2 + 1 = 9 marks)

END OF TEST